431 Quiz 1 for Fall 2026

Deadline: Wednesday 2026-10-14 at Noon

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This is the end of the Quiz. Congratulations! **34**

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Instructions for Students

There are **36** questions on this Quiz and this PDF is **36** pages long. Be sure you have all **36** pages[1]. It is to your advantage to answer all **36** questions. Your score is based on correct responses and some partial credit. There's no chance a blank response will be correct, and a guess might be, so you should definitely answer all questions.

This is an open book, open notes quiz. You are welcome to consult the materials provided on the course website and that we've been reading in the class, but you are not allowed to post the questions online, use some sort of AI to help you, or discuss the questions on this quiz with anyone other than Dr. Love (not even the teaching assistants.) You will be required to complete a short affirmation that you have obeyed these rules as part of submitting the Quiz.

[1]The last **3** pages are just session information.

0.1 The Google Form Answer Sheet

All of your answers should be placed in the Google Form Answer Sheet, located at...

https://tinyurl.com/431-2026-quiz1-form

All of your answers must be submitted through the Google Form by noon on Wednesday 2026-10-14, without exception. The form will close at 12:30 PM on that date, and no extensions will be available, so do not wait until late in the morning on Wednesday to submit your work. We will only accept responses through the Google Form.

The Google Form contains places to provide your responses to each question, and a final affirmation where you'll type in your name to tell us that you followed the rules for the Quiz. You must complete that affirmation before you can submit your responses. When you submit your results (in the same way you submit a Minute Paper) you will receive an email copy of your submission, with a link that will allow you to edit your work.

If you wish to work on some of the quiz and then return later, you can do this by [1] completing the final question (the affirmation) which asks you to type in your full name, and then [2] submitting the quiz. You will then receive a link at your CWRU email which will allow you to return to the quiz as often as you like without losing your progress.

0.2 The Data Sets

I have provided **six** data sets (called **q01.xlsx, q14.xlsx, q16a.csv, q16b.csv, q23.csv** and **q35.csv**) that are mentioned in the Quiz. These can either be downloaded from our 431-data page, or found in our Shared Drive in the Quiz 1 folder's data sub-folder.

0.3 Getting Help

If you need clarification on a Quiz question, you have exactly two ways of getting help:

- Ask your quiz question via email to **431-help at case dot edu**.
- Ask your quiz question in person during Class 15 on 2026-10-13.

During the Quiz period (5 PM 2026-10-09 through 12:30 PM 2026-10-14) we will not answer questions about the Quiz in TA office hours, or in any other way. Instead, I will respond to questions sent before 9 AM on 2026-10-14 in a timely fashion. Some cautions:

- Specific questions are more likely to get helpful answers.
- I will not review your code or your English for you.
- I will not tell you if your answer is correct, or if it is complete.
- I will email all students if we find an error in the Quiz that needs fixing.

0.4 When Should I Ask For Help?

We recommend the following process.

- If you encounter a tough question, skip it, and build up your confidence by tackling other questions.
- When you return to the tough question, spend no more than 10-15 minutes on it. If you still don't have it, take a break (not just to do other questions) but an actual break.
- When you return to the question, it may be much clearer to you. If so, great. If not, spend 5-10 minutes on it, at most, and if you are still stuck, ask Dr. Love a question via email.
- This is not to say that you cannot ask us sooner than this, but you should **never, ever** spend more than 20 minutes on any question without asking for help.

If you are stuck, ASK A QUESTION

You should **NEVER** spend more than 20 minutes on any question without asking Dr. Love for help. Just email him at **431-help AT CASE DOT EDU**[2].

0.5 Scoring the Quiz

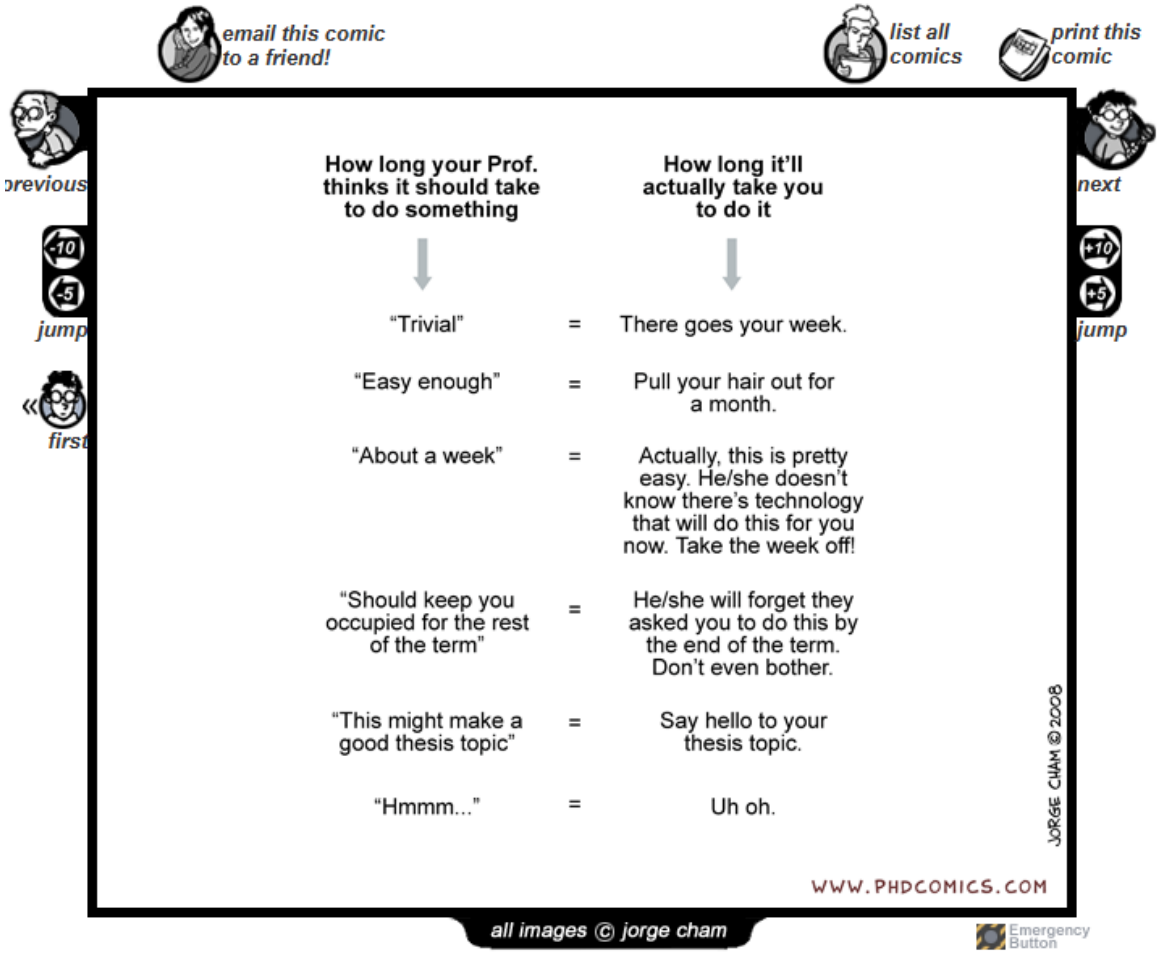
The questions are worth **3** points each yielding a total of **108** points. The questions are not in any particular order, and range in difficulty from "things I expect everyone to get right" to "things that are deliberately tricky".

[2] If you ignore this advice, that's your decision, but you'll have to live with the consequences. If you're taking more than 20 minutes to answer any of these questions, then you're very likely to be misunderstanding something.

0.6 How long should this take?

The Quiz is meant to take 6-7 hours. I expect most students will take 4-9 hours, and some will take as little as 3 or as many as 10. Some questions will take more time than others.

On the other hand...



Source: https://phdcomics.com/comics/archive.php?comicid=1093

0.7 Writing Code into the Google Form

Occasionally, we ask you to provide a single line of code. If not otherwise specified, a single line of code in response can contain **at most** two pipes, although you may or may not need the pipe in any particular setting. Note that I exclusively used the `|>` pipe, and not the `%>%` pipe, in developing this Quiz, but you may use either in your responses.

Do not include the `library` command at any time for your responses on the Quiz that ask for code. Assume in all questions that all packages listed below are loaded in R.

0.8 R Packages and Love-431.R script

Here is a list of packages in R. This **doesn't mean you need to use all of these packages**, and it also doesn't mean that I actually used all of them in building the Quiz and its answer sketch. It means only that I didn't use any other packages, other than **xfun** for session information, so you can do the entire Quiz without using any other packages.

I have also provided this listing to you at this link, so you can easily copy and paste it into your own RStudio session.

```
knitr::opts_chunk$set(comment = NA)

library(here)
source(here("data/Love-431.R"))

library(car)
library(DescTools)
library(Epi)
library(glue)
library(infer)
library(janitor)
library(knitr)
library(MKinfer)
library(naniar)
library(palmerpenguins)
library(patchwork)
library(readxl)
library(rstanarm)
library(easystats)
library(tidyverse)

theme_set(theme_bw())
```

> **Tip**
>
> Statistics is a "getting the details right" business. Read each question carefully, to be sure you get the details right in your response.

1 Question 1 (Q01)

You will be analyzing data from the **q01** data set in Questions 1-7.

The **q01** data contain (simulated) information from a real study for subjects presenting with urinary stones. We will assume that the data in **q01** describe a random sample of adult subjects presenting with urinary stones to a series of health care systems in the US. Variables include:

- `subject`: a subject identifying code
- `location`: stone location (bladder, ureter or kidney)
- `infectious`: yes if the stone composition was infectious[3]
- `urinepH`: urine pH as measured[4] from the subject's sample
- `diabetes`: whether the subject has an existing diagnosis of diabetes

Provide one line of R code (using at most three pipes) to display how stone location is related to infectious status, across all subjects in the **q01** tibble. Assume you've loaded the **q01** tibble, and (as for all questions) the R packages listed in the instructions.

2 Question 2 (Q02)

In Question 2, we restrict the sample discussed in Question 1 to those subjects with an existing diabetes diagnosis. Among those subjects, we ask you to do the following.

Specify the relative risk of a stone being infectious if it is located in the ureter (among subjects with a prior diagnosis of diabetes.) Then provide a 90% confidence interval around the relative risk. Only the numbers are required here.

[3] Infectious stones were those which contained at least one of carbonate apatite, ammonium urate or magnesium ammonium phosphate.

[4] Urinary pH is measured in points, with a normal range of 4.5 to 8.0 and with higher values indicating more alkaline urine, and lower values more acidic urine.

3 Question 3 (Q03)

Generate an appropriate visualization of all 1067 observations in the Question 1 data to compare the urine pH for subjects who had infectious vs. non-infectious stones. Which of the following statements best describes the patterns you see in the subjects with **infectious** stones[5]?

a. The urine pH levels of these subjects follow an approximately Normal distribution.
b. The urine pH levels of these subjects follow a symmetric, but clearly not Normal distribution.
c. The urine pH levels of these subjects follow a right-skewed distribution.
d. The urine pH levels of these subjects follow a left-skewed distribution.
e. None of answers a-d match the data I see.

4 Question 4 (Q04)

Use the visualization you created for Question 3 to answer the following. Which of these statements best describes the patterns you see in the subjects with **non-infectious** stones?

a. The urine pH levels of these subjects follow an approximately Normal distribution.
b. The urine pH levels of these subjects follow a symmetric, but clearly not Normal distribution.
c. The urine pH levels of these subjects follow a right-skewed distribution.
d. The urine pH levels of these subjects follow a left-skewed distribution.
e. None of answers a-d match the data I see.

5 Question 5 (Q05)

Provide the R code you would use to generate an appropriate point estimate, and 90% confidence interval, in order to compare the mean urine pH level of subjects with infectious stones to subjects with non-infectious stones across the 1,067 subjects in the **q01** data. Restrict yourself to at most two lines of R code.

6 Question 6 (Q06)

In a complete sentence or two, explain why you selected the approach you did in Question 5. Be sure to comment on whether the data describe paired or independent samples as part of your response, and justify that conclusion.

[5]Note that Question 4 asks about the subjects with non-infectious stones.

7 Question 7 (Q07)

In a complete sentence or two, provide the point estimate and 90% confidence interval you obtain using the code you generated in Question 5, and describe what they mean in the context of the study.

8 Question 8 (Q08)

You are working with the **starwars** tibble available in the `dplyr` package. How many characters with a **skin_color** that is `light` are **not** from the **homeworld** 'Naboo'?

Note that those with no homeworld information provided should not be included in your count.

9 Question 9 (Q09)

Again, you are working with the starwars tibble mentioned in Question 8. Now, you want to identify characters who are either 'Droid' or 'Human', and then for each of these species, determine the number of distinct `eye_color` values they possess.

Provide a single line of R code (using at most three pipes) that will produce the count of unique `eye_color` values for each of the selected species.

10 Question 10 (Q10)

You are building a linear model to predict speed based on height of a roller coaster. After fitting a linear model, you decide to check assumptions using `check_model()` with `detrend = TRUE`. If the "Normality of Residuals" plot, showed points that curved upward away from the line at both the lower and upper ends, but with inner points closer to the line, this pattern would typically indicate which of the following characteristics about the residuals?

a. Right-skewed residuals.
b. Left-skewed residuals.
c. Heavy-tailed residuals (more outliers than expected in a Normal distribution).
d. Light-tailed residuals (fewer outliers than expected in a Normal distribution).
e. (essentially) Normally distributed residuals.
f. None of the above.

11 Question 11 (Q11)

You are analyzing the **penguins** dataset (available in the **palmerpenguins** package) and want to find the number of penguins with missing sex information. Which of the following lines of code will provide this information? Assume the palmerpenguins package is loaded and the penguins tibble within that package is available.

(Check all of the correct responses.)

a. summary(penguins)
b. penguins |> count(sex)
c. tabyl(penguins$sex)
d. describe_distribution(penguins, include_factors = TRUE)
e. penguins |> select(sex) |> table()
f. None of responses a-e are correct.

12 Question 12 (Q12)

You have fit an OLS linear model and obtained the diagnostic plots from `check_model()`. The "Linearity" plot (residuals vs. fitted values) shows a distinct U-shaped curve. What does this pattern primarily suggest about your model?

a. The residuals are not Normally distributed.
b. There is a problem with heteroscedasticity (non-constant variance).
c. The relationship between the predictor(s) and the outcome is non-linear.
d. There are influential outliers affecting the model.
e. The model has high collinearity among predictors.
f. None of the above.

13 Question 13. (Q13)

In the PPDAC framework, what are the primary dangers of a poorly or vaguely defined "Problem" stage? (CHECK ALL THAT APPLY)

a. Failing to randomize the study design properly due to sudden budget constraints during execution of the study.
b. It forces the analysis to include detailed checks of model assumptions.
c. Encountering severe missing values and data corruption during the initial data cleaning phase.
d. It leads to answering a statistically rigorous question that completely misses the real-world context.
e. Collecting data that is entirely irrelevant or insufficient to address the actual underlying question of interest.
f. It leads to hiding confidence intervals so the public is not confused by statistical jargon.
g. None of the above.

14 Question 14 (Q14)

The data available in the `q14.xlsx` file provided with the quiz contains, among other things, a variable called `lines`, which is the number of Quarto lines of code included in the final submission for 42 projects submitted in Fall 2023. We're going to create three confidence intervals based on this sample, which has sample mean = 768.4 lines and sample median = 703 lines of code.

a. Option A is to set a seed to 431, then obtain a 95% confidence interval for the population *mean* number of lines of code, using the `boot.t.test()` function from the MKinfer package, with 2000 bootstrap replications.

b. Option B is to obtain a 95% confidence interval for the population *mean* number of lines of code, using an ordinary least squares linear model and the `model_parameters()` function.

c. Option C is to again set a seed to 431, and then obtain a 95% confidence interval for the population *median* number of lines of code, this time using a bootstrap percentile interval obtained using the **infer** package and code like that shown in section 5.5.2 of Dr. Love's course book.

Columns: Option A, Option B, Option C

Rows:

1. Which is the best choice if the data were randomly sampled from a population that is described well by the Normal distribution?

2. Which produces the confidence interval with the smallest width?
3. Which produces the largest upper bound for its confidence interval?
4. Which is symmetric around its point estimate?

15 Question 15 (Q15)

Consider again the data from Question 14, but here we are developing a model to predict the number of `lines` of code using the number of `characters` they used in writing the Project A reflection statement.

Which of the following transformations of `lines` does a Box-Cox approach suggest will be the best option for building such a model?

a. No transformation
b. Squaring the `lines`
c. The inverse of `lines`
d. The natural logarithm of `lines`
e. The square root of `lines`
f. The squared inverse of `lines`

16 Question 16 (Q16)

For Questions 16-19, I have provided you with two data sets, called **q16a.csv** and **q16b.csv**. The two data sets contain the same information, but arranged differently. In each case, what you have are the infant mortality (birth to 1 year old) data for 201 countries, measured in deaths per 1000 live births for two periods of time: the years between 1995 and 2000 and the years between 2015 and 2020. Our goal is to compare the average infant mortality level between the earlier and later periods. Ingest the data into R in whatever manner appeals to you, and then build a visualization that allows you to assess the Normality of the data in a way that makes sense in context.

Which of the following statements best matches your conclusions?

a. The data describe two independent samples, and each follows an (approximately) Normal distribution.
b. The data describe two independent samples, and at least one of the two samples is symmetric, but not Normally distributed.
c. The data describe two independent samples, and at least one of the two samples shows meaningful skew.
d. The data describe two paired samples, and the paired differences follow an (approximately) Normal distribution.
e. The data describe two paired samples, and the paired differences follow a symmetric but heavy-tailed distribution.
f. The data describe two paired samples, and the paired differences follow a symmetric but light-tailed distribution.
g. The data describe two paired samples, and the paired differences follow a skewed distribution.
h. None of these statements are correct.

17 Question 17 (Q17)

Complete an appropriate comparison of the population mean infant mortality rates associated with the two time periods described in Question 16, and specify the point estimate and a 95% confidence interval around that point estimate, rounding your estimates of the newer - older difference to two decimal places. (Here, we're just asking you to specify the numbers.)

18 Question 18 (Q18)

Provide an appropriate description of the meaning of the confidence interval you specified in Question 17, in one to three sentences.

19 Question 19 (Q19)

In Question 16, you chose to use either paired or independent samples. Justify that choice, in one to three sentences.

20 Question 20 (Q20)

You are presented with a "Posterior Predictive Check" plot from a linear model (`check_model()` output) that predicts an outcome (called `out`) using a quantitative predictor variable called `pred`. The plot shows that the observed data density curve has two distinct peaks, while the model-predicted data curves are unimodal and fall substantially below the observed curve in the peaks and above it in the trough. What does this pattern indicate about your model?

(Check all of the correct responses.)

a. The underlying `out` data are not linearly related to `pred`
b. The `out` data are highly skewed and require a power transformation.
c. The model for `out` has a problem with non-constant variance.
d. The `pred` variable is not highly correlated with `out`.
e. None of responses a-d are correct.

21 Question 21 (Q21)

Way back in Class 1, Professor Love mentioned that Artificial Intelligence tools should primarily be used for "things that they already know how to do well" when studying. What is the underlying pedagogical reason for this advice?

a. To ensure students always get the correct answer without effort.
b. To prevent students from engaging with material that might teach them something new.
c. To encourage students to rely solely on AI for problem-solving.
d. To prioritize learning by doing tasks that provide valuable learning experience.
e. To discourage the use of AI in all academic contexts.

22 Question 22 (Q22)

What does the `set.seed()` function do in R, and why is it important for reproducible research, especially when using functions that involve random number generation (e.g., bootstrap procedures, `rnorm()`, `stan_glm()`)?

Answer this question in a sentence or two.

23 Question 23 (Q23)

The **q23** data provided to you is used in Questions 23-25. The data contains a subject identifying code, and two main variables:

- `hosp_costs`: the costs (in US $) incurred by the hospital for the patient's care (prior to payment by insurance or the patient)
- `satisfaction`: the patient's satisfaction rating regarding the quality of care they received from nurses during their stay (Excellent, VeryGood, Good, Fair or Poor).

The sample was created by taking a random sample of the same number of patients (101) from each of the five `satisfaction` groups from among patients seen in the past year.

For question 23, build an appropriate analysis of variance table and specify (a) the eta-squared value, expressed as a proportion rounded to three decimal places, and (b) the mean square of the residuals, rounded to the nearest integer.

24 Question 24 (Q24)

Which of the following statements describe the results of a Tukey HSD comparison for the ANOVA you built in Question 23, which requires a 95% family-wise confidence level for the complete set of pairwise comparisons? Hint: interpret "A is meaningfully smaller than B" in these responses to mean that the difference between A and B yields a Tukey HSD confidence interval that is entirely negative. (Check all of the true statements.)

a. The Excellent group mean is meaningfully smaller than all other groups.
b. The Good and Very Good group means appear meaningfully different from each other.
c. The Fair and Poor group means appear meaningfully larger than the means of the other three groups.
d. The Poor group mean appears meaningfully larger than the Fair mean.
e. None of the group means appear meaningfully different from any other.

25 Question 25 (Q25)

Which of the following `check_model()` plots show meaningful problems with assumptions for the ANOVA model you built in Question 23?

(Check all of the plots which show such problems.)

a. The posterior predictive check.
b. The linearity check.
c. The homogeneity of variance check.
d. The influential observations check.
e. The normality of residuals check.
f. None of these plots show meaningful problems.

26 Question 26 (Q26)

In reviewing descriptive statistics for a continuous variable, you observe that the mean is substantially higher than the median, and the distribution displays a long tail to the right. Which pair of summary statistics would generally be the most appropriate to describe the center and spread of this variable?

a. The mean and standard deviation
b. The median and median absolute deviation (MAD)
c. The mean and interquartile range
d. The mode and range
e. All are equally useful.

Output for Questions 27-32

Questions 27-32 use the output appearing on pages 18-27.

I selected a random sample of 1,000 adults residing in the US from a large survey. The information available includes each subject's age (in years), sex and body mass index (measured in kg/m^2.) The data are available to me (but **not** to you) in a tibble called **q27.Rds**.

Output part 1 (ingesting the data)

Here are the first two and last two rows of that tibble.

```
q27 <- read_rds("data/q27.Rds")

head(q27, 2)
```

```
# A tibble: 2 x 4
  subj  sex      age   bmi
  <chr> <fct>  <dbl> <dbl>
1 1001  Male      27  23.4
2 1002  Female    50  33.9
```

```
tail(q27, 2)
```

```
# A tibble: 2 x 4
  subj  sex      age   bmi
  <chr> <fct>  <dbl> <dbl>
1 1999  Female    57  46
2 2000  Female    24  22.9
```

Here is a brief summary of that tibble.

```
summary(q27)
```

```
        subj          sex           age              bmi
 Length   :1000   Male  :497   Min.   :20.00   Min.   :15.80
 N.unique :1000   Female:503   1st Qu.:28.00   1st Qu.:23.77
 N.blank  :   0                Median :38.00   Median :27.30
 Min.nchar:   4                Mean   :38.61   Mean   :28.72
 Max.nchar:   4                3rd Qu.:48.25   3rd Qu.:32.30
                               Max.   :59.00   Max.   :80.60
```

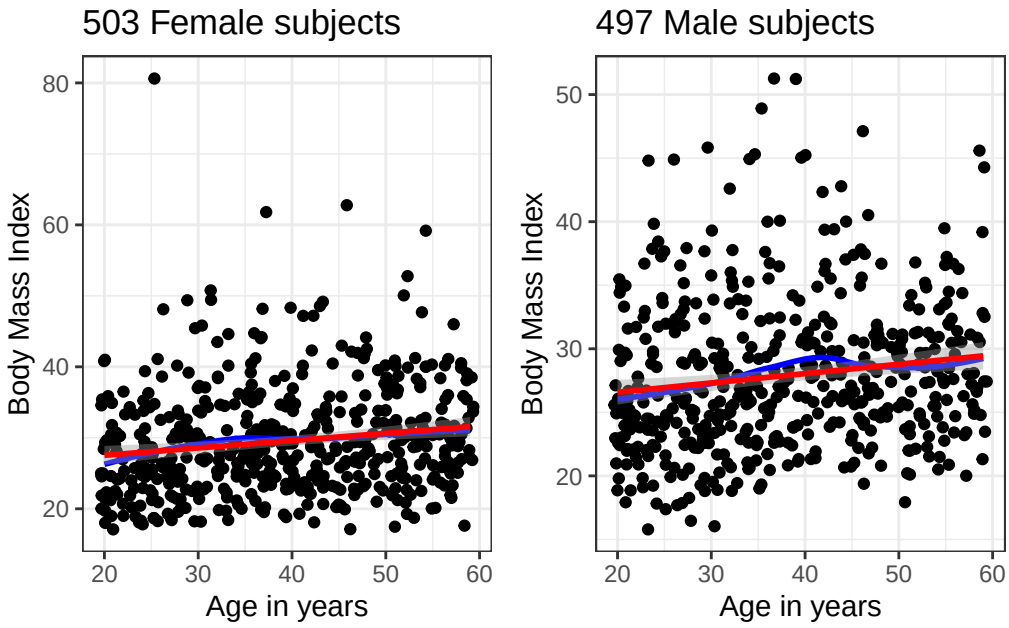
Output part 2 (an important pair of plots)

```
dat27a <- q27 |> filter(sex == "Female")
dat27b <- q27 |> filter(sex == "Male")

plota <- ggplot(dat27a, aes(x = age, y = bmi)) +
    geom_jitter() +
    geom_smooth(method = "loess", col = "blue", se = FALSE, formula = y ~ x) +
    geom_smooth(method = "lm", col = "red", se = TRUE, formula = y ~ x) +
    labs(x = "Age in years", y = "Body Mass Index",
         title = glue(nrow(dat27a), " Female subjects"))

plotb <- ggplot(dat27b, aes(x = age, y = bmi)) +
    geom_jitter() +
    geom_smooth(method = "loess", col = "blue", se = FALSE, formula = y ~ x) +
    geom_smooth(method = "lm", col = "red", se = TRUE, formula = y ~ x) +
    labs(x = "Age in years", y = "Body Mass Index",
         title = glue(nrow(dat27b), " Male subjects"))

plota + plotb
```



Output part 3 (fitting two models)

```
m27a <- lm(bmi ~ age, data = dat27a)
model_parameters(m27a, pretty_names = FALSE, ci = 0.90, digits = 3)
```

```
Parameter   | Coefficient |    SE |           90% CI | t(501) |      p
-------------------------------------------------------------------------
(Intercept) |      25.469 | 1.230 | [23.441, 27.496] | 20.700 | < .001
age         |       0.103 | 0.030 | [ 0.054,  0.153] |  3.434 | < .001

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_performance(m27a)
```

```
# Indices of model performance

AIC    |   AICc |    BIC |    R2 | R2 (adj.) |  RMSE | Sigma
-----------------------------------------------------------------
3500.2 | 3500.2 | 3512.9 | 0.023 |     0.021 | 7.802 | 7.818
```

```
m27b <- lm(bmi ~ age, data = dat27b)
model_parameters(m27b, pretty_names = FALSE, ci = 0.90, digits = 3)
```

```
Parameter   | Coefficient |    SE |           90% CI | t(495) |      p
-------------------------------------------------------------------------
(Intercept) |      25.108 | 0.915 | [23.600, 26.615] | 27.452 | < .001
age         |       0.074 | 0.023 | [ 0.036,  0.111] |  3.198 | 0.001

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

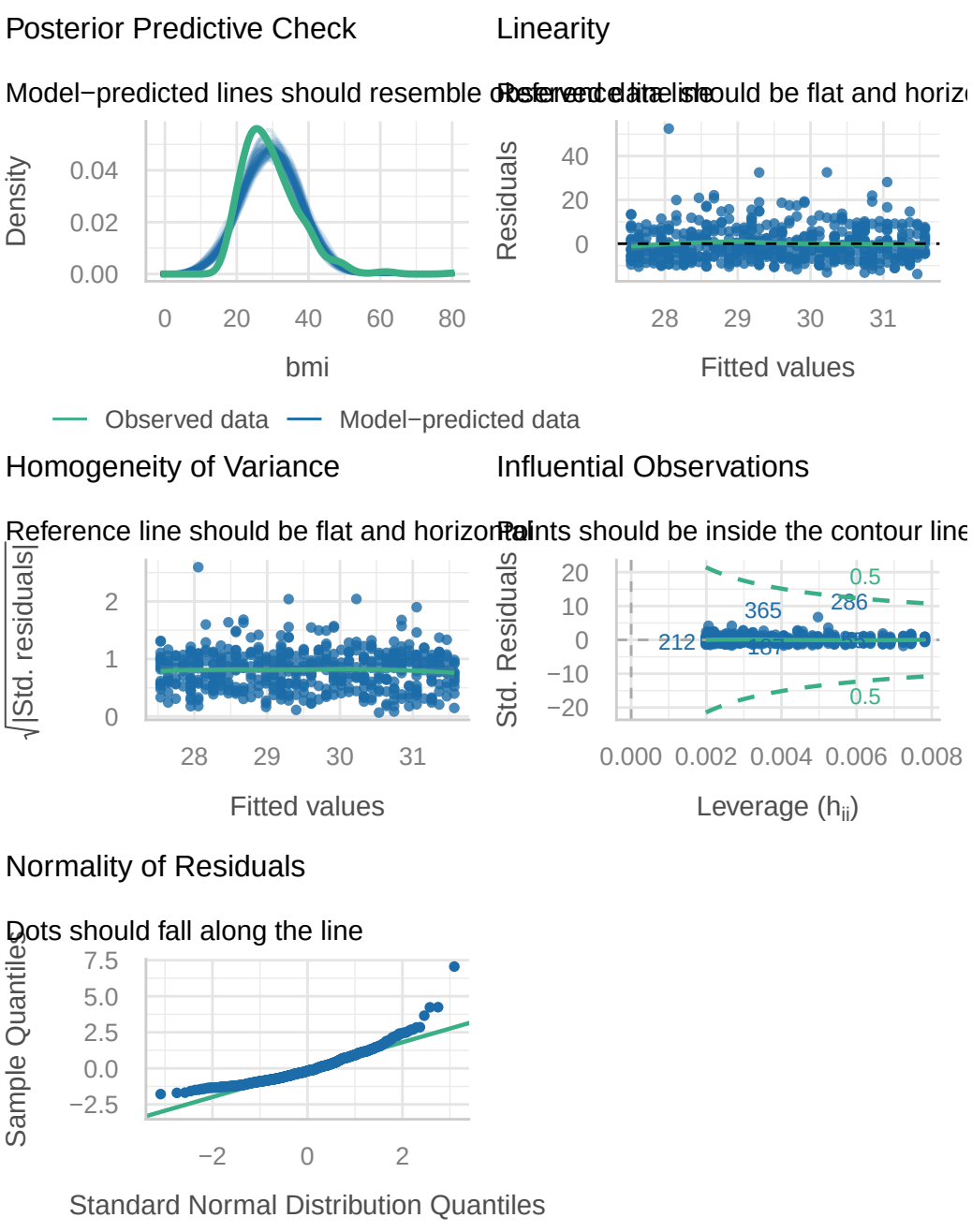
```
model_performance(m27b)
```

```
# Indices of model performance

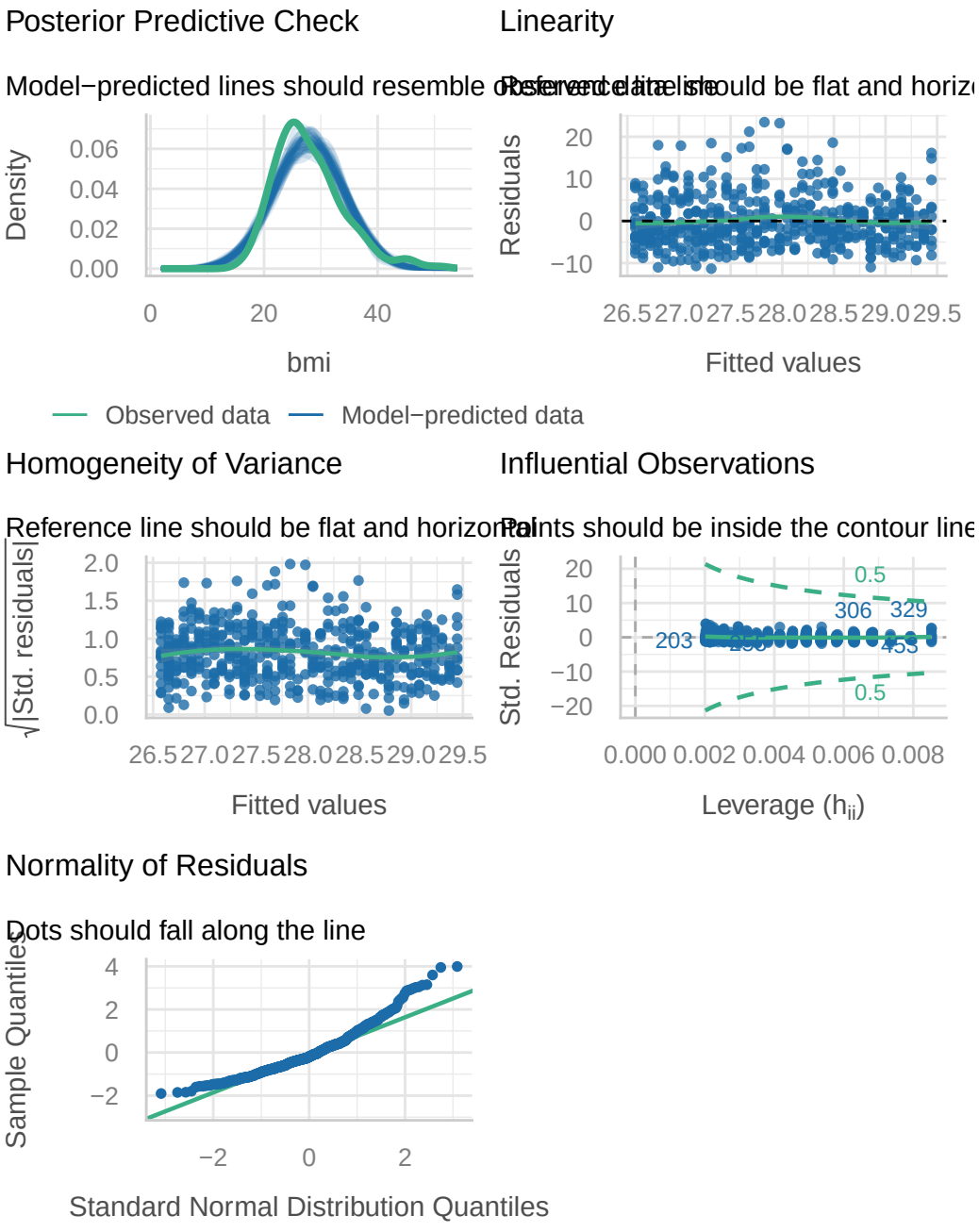
AIC    |   AICc |    BIC |    R2 | R2 (adj.) |  RMSE | Sigma
-----------------------------------------------------------------
3190.9 | 3191.0 | 3203.5 | 0.020 |     0.018 | 5.961 | 5.973
```

Output part 4 (checking assumptions for two models)

```
check_model(m27a, detrend = FALSE)
```

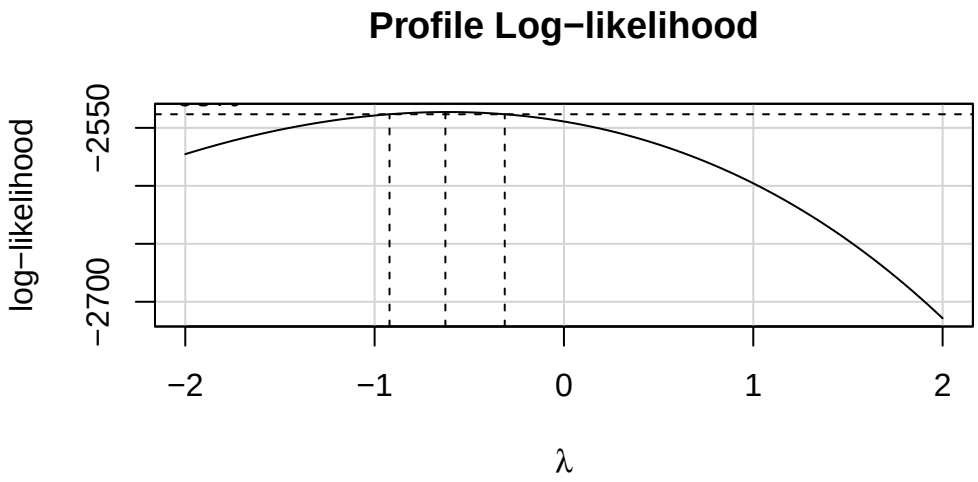


```
check_model(m27b, detrend = FALSE)
```

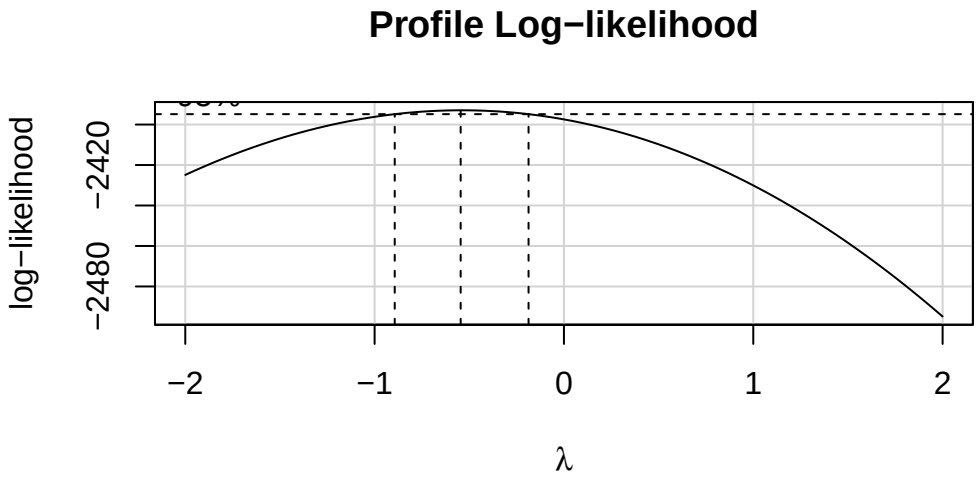


Output part 5 (Box-Cox plots)

```
boxCox(m27a)
```




```
boxCox(m27b)
```



Output part 6 (Transformation 1)

```
dat27a <- dat27a |> mutate(bmi_tr = 1000/bmi)
head(dat27a, 2)
```

```
# A tibble: 2 x 5
  subj  sex      age   bmi bmi_tr
  <chr> <fct>  <dbl> <dbl>  <dbl>
1 1002  Female    50  33.9   29.5
2 1003  Female    39  34.7   28.8
```

```
m27a_tr <- lm(bmi_tr ~ age, data = dat27a)
model_parameters(m27a_tr, pretty_names = FALSE, ci = 0.90)
```

```
Parameter   | Coefficient |   SE |         90% CI | t(501) |      p
-------------------------------------------------------------------
(Intercept) |       41.62 | 1.34 | [39.41, 43.83] |  31.05 | < .001
age         |       -0.14 | 0.03 | [-0.20, -0.09] |  -4.33 | < .001

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_performance(m27a_tr)
```

```
# Indices of model performance

AIC    |   AICc |    BIC |    R2 | R2 (adj.) |  RMSE | Sigma
---------------------------------------------------------------
3586.3 | 3586.4 | 3599.0 | 0.036 |     0.034 | 8.500 | 8.517
```

Output part 7 (Transformation 2)

```
dat27b <- dat27b |> mutate(bmi_tr = 100/sqrt(bmi))
head(dat27b, 2)
```

```
# A tibble: 2 x 5
  subj  sex     age   bmi bmi_tr
  <chr> <fct> <dbl> <dbl>  <dbl>
1 1001  Male     27  23.4   20.7
2 1004  Male     37  22.7   21.0
```

```
set.seed(43127)
m27b_tr <- stan_glm(bmi_tr ~ age, data = dat27b, refresh = 0)
model_parameters(m27b_tr, pretty_names = FALSE, ci = 0.90)
```

```
Parameter   | Median |         90% CI |   pd |  Rhat | ESS (tail) |                  Prior
-------------------------------------------------------------------------------------------
(Intercept) |  20.36 | [19.87, 20.86] | 100% | 1.000 |       2612 | Normal (19.24 +- 4.93)
age         |  -0.03 | [-0.04, -0.02] | 100% | 1.001 |       2890 |  Normal (0.00 +- 0.42)

Uncertainty intervals (equal-tailed) computed using a MCMC distribution
  approximation.
```

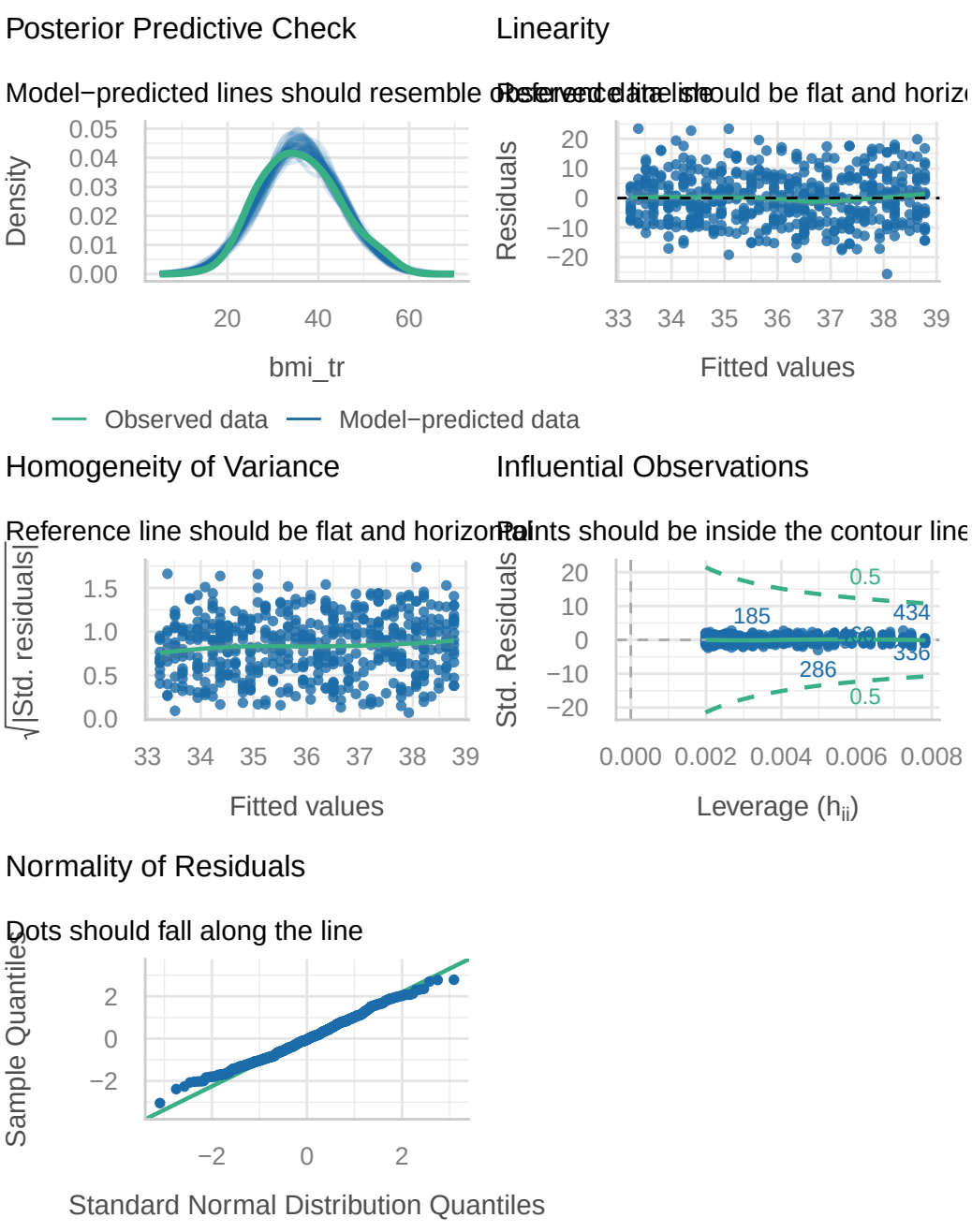
```
model_performance(m27b_tr)
```

```
# Indices of model performance

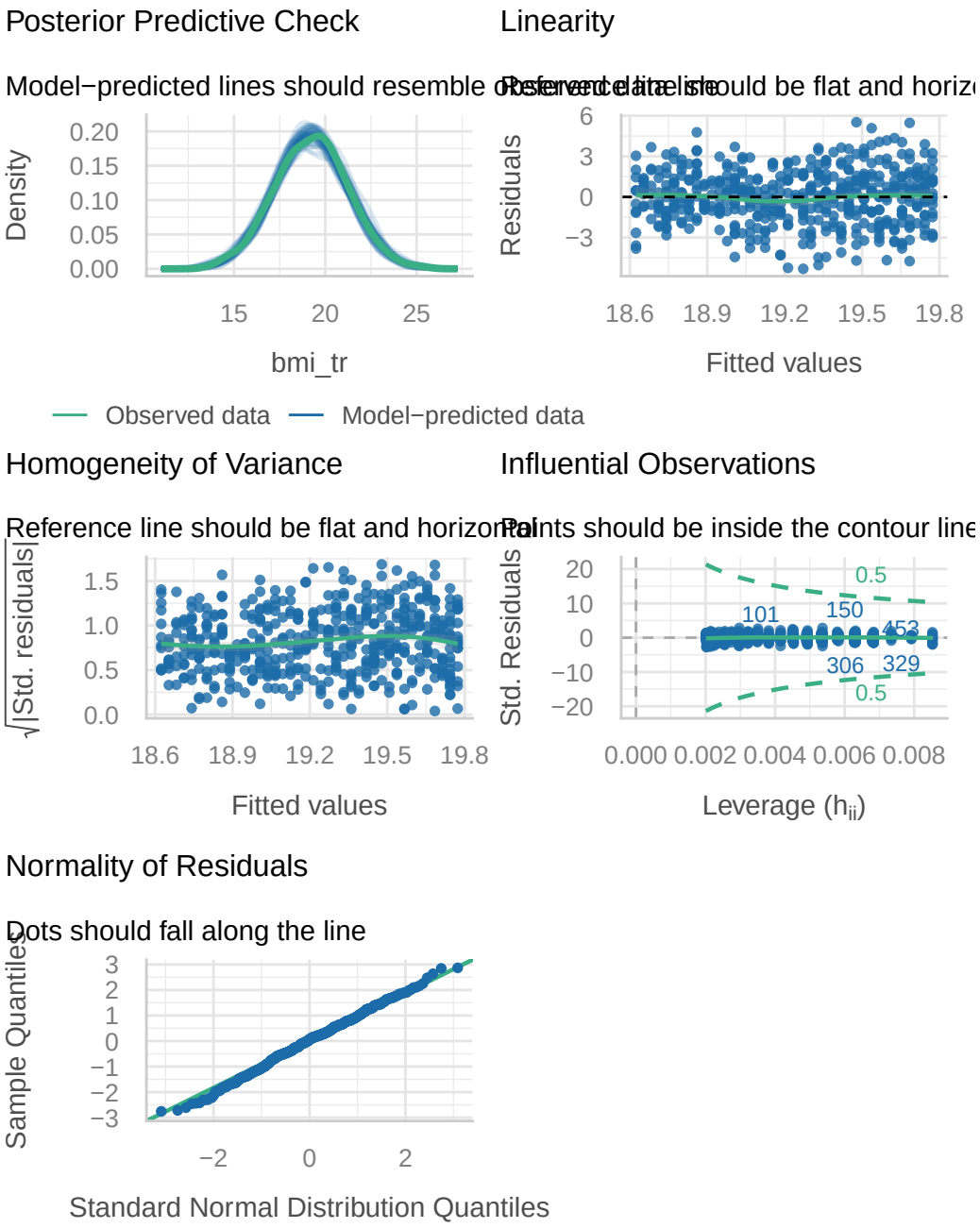
ELPD      | ELPD_SE |  LOOIC | LOOIC_SE |   WAIC |    R2 | R2 (adj.) |  RMSE | Sigma
-------------------------------------------------------------------------------------
-1037.272 |  15.411 | 2074.5 |   30.822 | 2074.5 | 0.030 |     0.025 | 1.939 | 1.945
```

Output part 8 (Checking assumptions for new models)

```
check_model(m27a_tr, detrend = FALSE)
```



```
check_model(m27b_tr, detrend = FALSE)
```



This is the end of the output for Questions 27-32.

27 Question 27 (Q27)

Refer to the Output for Questions 27-32 provided on pages 18-27.

Which of the following statements are true? (Check all of the true statements.)

a. There is no clear indication of non-linearity in the scatterplot of the 503 female subjects.
b. There is no clear indication of non-linearity in the scatterplot of the 497 male subjects.
c. The point estimate from model m27a for a 38-year old female's BMI is larger than the point estimate from model m27b for a 38-year old male's BMI.
d. Model m27a will provide equally accurate results when predicting a female subject's BMI if she is 25 years old as it will if she is 75 years old.
e. None of statements a-d are true.

28 Question 28 (Q28)

Which of the following regression model concerns appear seriously violated?

(Check any indications of serious violations)

- Column 1: for model 27a?
- Column 2: for model 27b?

a. The assumption of linearity.
b. The assumption of constant variance (homoscedasticity.)
c. The assumption of Normality.
d. The notion that none of the points should be particularly influential.
e. The R-square value is too high to be useful to us.
f. No serious violations are present.

29 Question 29 (Q29)

Refer to the Output for Questions 27-32 provided on pages 18-27.

Which of the following statements are true? (Check all of the true statements.)

a. Model m27a_tr uses the choice of transformation that is suggested by the appropriate Box-Cox plot.
b. Model m27b_tr uses the choice of transformation that is suggested by the appropriate Box-Cox plot.
c. The R-square for model m27a and model m27a_tr are comparable to each other, because they describe the same sample of 503 female subjects.
d. Since the Sigma estimate for m27a_tr (8.517) is greater than the Sigma estimate (1.945) for m27b_tr, the m27a model must not provide as precise a prediction of BMI.
e. None of statements a-d are true.

30 Question 30 (Q30)

What is the predicted BMI for a 38 year old male subject, using model m27b_tr? Please round your answer to two decimal places.

31 Question 31 (Q31)

Which of the following plots show meaningful problems?

(Check all plots which show meaningful problems.)

- Column 1: our transformed model `m27a_tr`
- Column 2: our transformed model `m27b_tr`

Rows:

a. The posterior predictive check
b. The linearity plot
c. The homogeneity of variance plot
d. The influential observations plot
e. The Normality of residuals plot
f. None of the plots specified in a-e.

32 Question 32 (Q32)

Refer to the Output for Questions 27-32 provided on pages 18-27.

Specify (as best you can given the information provided to you) the age, sex and BMI of the subject with the largest BMI across the 1000 subjects in the q27 data.

33 Question 33 (Q33)

Bootstrapping a sample can be used in many settings. Which of the following statements are true about the bootstrap? (Check all of the true statements.)

a. When we use the bootstrap, we create new data sets of the same size, by sampling the original data, without replacement.
b. Sample statistics calculated from bootstrap re-samples of reasonably large but skewed data look more Normally distributed, in general, than the original data.
c. Uncertainty intervals built using the bootstrap do not require strong distributional assumptions about the underlying data.
d. Bootstrap-based confidence intervals for the sample mean are less wide than those based on the t distribution.
e. A bootstrap confidence interval relies heavily on the assumption that our data re-samples are drawn at random.
f. None of statements a-e are true.

34 Question 34 (Q34)

Why is relying solely on "statistical significance" (e.g., $p < 0.05$) problematic for drawing scientific conclusions? (Check all of the correct statements.)

a. It encourages overly complex statistical models.
b. It transforms uncertainty into rigid "yes/no" answers, hindering a nuanced understanding of data.
c. It discourages the use of uncertainty intervals.
d. It is only applicable to observational studies, not experiments.
e. It always leads to false positive findings.

Output for Questions 35-36

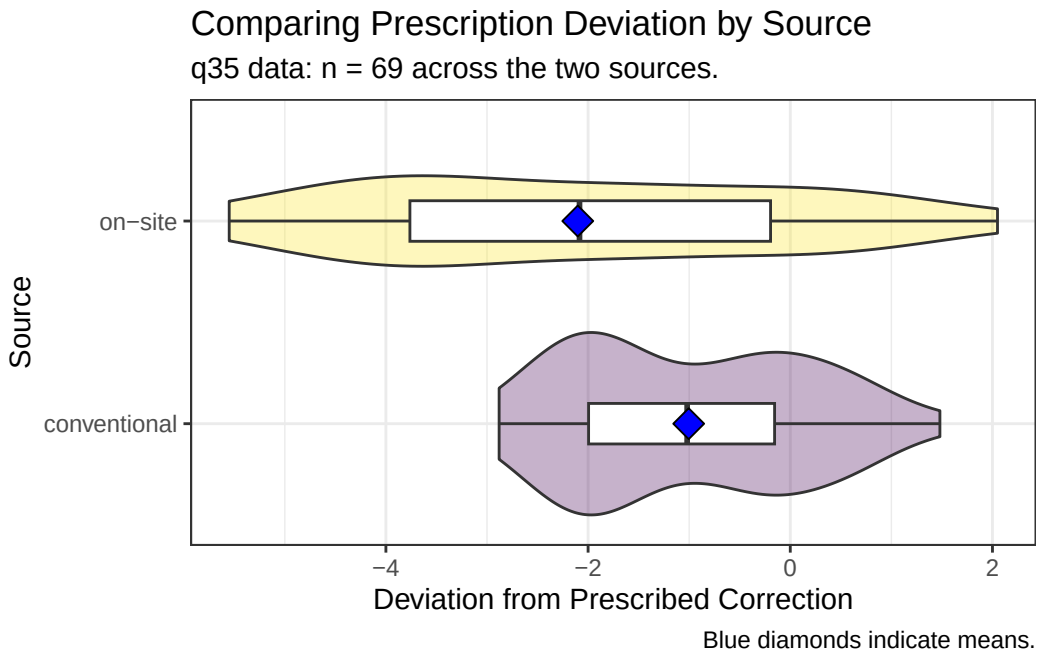
Recently, a number of opticians established on-site laboratories for preparing prescription eyeglasses. These labs provide more rapid service than conventional off-premises labs. Conventional opticians have questioned the accuracy of on-site labs. As a test, eyeglasses prescribed for nearsightedness were prepared by both types of labs. The glasses were then evaluated by very accurate devices that determine the percentage deviation from the prescribed correction. A minus sign indicates that the actual correction is less than prescribed; a plus sign, more than prescribed.

The data used in this study are available to you in the `q35.csv` file.

```
q35 <- read_csv("data/q35.csv", show_col_types = FALSE)

q35 |> group_by(source) |> reframe(lovedist(deviation))
```

```
# A tibble: 2 x 11
  source           n  miss  mean    sd   med   mad   min   q25    q75   max
  <chr>        <int> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl>
1 conventional    43     0 -1.00  1.15 -1.02  1.41 -2.88 -2.00 -0.155  1.48
2 on-site         26     0 -2.10  2.12 -2.08  2.71 -5.55 -3.76 -0.195  2.05
```



Three analyses (labeled Analysis 1 - Analysis 3) using these data follow.

Question 35 Analysis 1

```
model_parameters(lm(deviation ~ source, data = q35), ci = 0.90)
```

```
Parameter        | Coefficient |   SE |         90% CI | t(67) |      p
-------------------------------------------------------------------------
(Intercept)      |       -1.00 | 0.24 | [-1.41, -0.60] | -4.16 | < .001
source [on-site] |       -1.10 | 0.39 | [-1.75, -0.44] | -2.79 | 0.007

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

Question 35 Analysis 2

```
set.seed(431035)
boot.t.test(deviation ~ source, data = q35, var.equal = FALSE,
            R = 2000, conf.level = 0.90)
```

```
    Bootstrap Welch Two Sample t-test

data:  deviation by source
number of bootstrap samples:  2000
bootstrap p-value = 0.03
bootstrap difference of means (SE) = 1.103535 (0.4420651)
90 percent bootstrap percentile confidence interval:
 0.3770367 1.7964960

Results without bootstrap:
t = 2.4341, df = 34.077, p-value = 0.02032
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 0.3350552 1.8594886
sample estimates:
mean in group conventional      mean in group on-site
                 -1.004651                  -2.101923
```

Question 35 Analysis 3

```
wilcox.test(deviation ~ source, data = q35, exact = F,
            conf.int = T, conf.level = 0.90)
```

```
    Wilcoxon rank sum test with continuity correction

data:  deviation by source
W = 734.5, p-value = 0.03023
alternative hypothesis: true location shift is not equal to 0
90 percent confidence interval:
 0.3200056 1.9899889
sample estimates:
difference in location
               1.29454
```

35 Question 35 (Q35)

Which of these 90% confidence intervals is our best choice for estimating the mean of the population conventional minus on-site difference in percentage deviation?

a. (0.60, 1.41)
b. (0.44, 1.75)
c. (0.38, 1.80)
d. (0.34, 1.86)
e. (0.32, 1.99)

36 Question 36 (Q36)

In a sentence or two, specify why you chose the confidence interval you selected in Question 35.

This is the end of the Quiz. Congratulations!

Session Information

```
xfun::session_info()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Locale:
  LC_COLLATE=English_United States.utf8
  LC_CTYPE=English_United States.utf8
  LC_MONETARY=English_United States.utf8
  LC_NUMERIC=C
  LC_TIME=English_United States.utf8

Package version:
  abind_1.4-8            ape_5.8.1              arrangements_1.1.10
  askpass_1.2.1          backports_1.5.1        base_4.6.1
  base64enc_0.1-6        bayesplot_1.16.0       bayestestR_0.19.0
  BH_1.90.0.1            bit_4.6.0              bit64_4.8.6
  blob_1.3.0             boot_1.3-32            broom_1.0.13
  bslib_0.12.0           cachem_1.1.0           callr_3.8.0
  car_3.1-5              carData_3.0-6          cellranger_1.1.0
  checkmate_2.3.4        class_7.3-24           cli_3.6.6
  clipr_0.8.1            cmprsk_2.2-12          coda_0.19-4.1
  codetools_0.2-20       colorspace_2.1.3       colourpicker_1.3.0
  commonmark_2.0.0       compiler_4.6.1         conflicted_1.2.0
  contfrac_1.1-12        correlation_0.8.8      cowplot_1.2.0
  cpp11_0.5.5            crayon_1.5.3           crosstalk_1.2.2
  curl_8.0.0             data.table_1.18.6.1    datasets_4.6.1
  datawizard_1.4.0       DBI_1.3.0              dbplyr_2.6.0
  Deriv_4.3.5            desc_1.4.3             DescTools_0.99.60
  deSolve_1.42           DHARMa_0.5.0           digest_0.6.39
  distributional_0.9.0   doBy_4.7.2             doParallel_1.0.17
  dplyr_1.2.1            DT_0.34.0              dtplyr_1.3.3
  dygraphs_1.1.1.6       e1071_1.7-17           easystats_0.7.6
  effectsize_1.0.3       elliptic_1.5-1         emmeans_2.0.4
  Epi_2.66               estimability_2.0.0     etm_1.1.2
```

evaluate_1.0.5 Exact_3.3 exactRankTests_0.8-37
expm_1.0-1 farver_2.1.2 fastmap_1.2.0
fontawesome_0.5.3 forcats_1.0.1 foreach_1.5.2
forecast_9.0.2 Formula_1.2-6 fracdiff_1.5.4
fs_2.1.0 gap_1.15.2 gap.datasets_0.0.6
gargle_1.6.1 generics_0.1.4 ggdist_3.3.3
ggplot2_4.0.3 ggrepel_0.9.8 ggridges_0.5.7
gld_2.6.8 glmnet_5.1 glue_1.8.1
gmp_0.7-5.1 googledrive_2.1.2 googlesheets4_1.1.2
graphics_4.6.1 grDevices_4.6.1 grid_4.6.1
gridExtra_2.3.1 gtable_0.3.6 gtools_3.9.5
haven_2.5.5 here_1.0.2 highr_0.12
hms_1.1.4 htmltools_0.5.9 htmlwidgets_1.6.4
httpuv_1.6.17 httr_1.4.9 hypergeo_1.2-14
ids_1.0.1 igraph_2.3.3 infer_1.1.0
inline_0.3.21 insight_1.5.4 isoband_0.3.0
iterators_1.0.14 janitor_2.2.1 jomo_2.7-6
jquerylib_0.1.4 jsonlite_2.0.0 knitr_1.52
labeling_0.4.3 later_1.4.8 lattice_0.23-1
lazyeval_0.2.3 lifecycle_1.0.5 litedown_0.11
lme4_2.0-6 lmom_3.3 lmtest_0.9.40
loo_2.10.1 lubridate_1.9.5 magrittr_2.0.5
markdown_2.0 MASS_7.3-66 Matrix_1.7-6
MatrixModels_0.5.4 matrixStats_1.5.0 memoise_2.0.1
methods_4.6.1 mgcv_1.9-4 mice_3.19.0
miceadds_3.20-10 mime_0.13 miniUI_0.1.2
minqa_1.2.8 mitml_0.4-5 mitools_2.7
MKdescr_0.9 MKinfer_1.4 modelbased_0.17.0
modelr_0.1.11 multcomp_1.4-32 mvtnorm_1.4-2
naniar_1.1.0 nlme_3.1-171 nloptr_2.2.1
nnet_7.3-21 norm_1.0.11.1 numDeriv_2016.8-1.1
openssl_2.4.2 ordinal_2026.7.26 otel_0.2.0
palmerpenguins_0.1.1 pan_2.0 parallel_4.6.1
parameters_0.29.3 patchwork_1.3.2 pbkrtest_0.5.5
performance_0.18.2 pillar_1.11.1 pkgbuild_1.4.8
pkgconfig_2.0.3 plotly_4.12.1 plyr_1.8.9
posterior_1.7.0 prettyunits_1.2.0 processx_3.9.0
progress_1.2.3 promises_1.5.0 proxy_0.4-29
ps_1.9.3 purrr_1.2.2 qgam_2.0.0
quadprog_1.5.8 quantreg_6.1 QuickJSR_1.11.0
R6_2.6.1 ragg_1.5.2 rappdirs_0.3.4
rbibutils_2.4.1 RColorBrewer_1.1-3 Rcpp_1.1.2
RcppArmadillo_15.6.0.1 RcppEigen_0.3.4.0.2 RcppParallel_6.2.1

Rdpack_2.6.6	readr_2.2.0	readxl_1.5.0.1
reformulas_0.4.4	rematch_2.0.0	rematch2_2.1.2
report_0.6.4	reprex_2.1.1	reshape2_1.4.5
rlang_1.3.0	rmarkdown_2.32	rootSolve_1.8.2.4
rpart_4.1.27	rprojroot_2.1.1	rstan_2.32.7
rstanarm_2.32.2	rstantools_2.7.1	rstudioapi_0.19.0
rvest_1.0.5	S7_0.2.2	sandwich_3.1-3
sass_0.4.10	scales_1.4.0	see_0.14.2
selectr_0.8.0	shape_1.4.6.1	shiny_1.14.0
shinyjs_2.1.1	shinystan_2.7.0	shinythemes_1.2.0
snakecase_0.11.1	sourcetools_0.1.7.2	SparseM_1.84.2
splines_4.6.1	StanHeaders_2.39.1	stats_4.6.1
stats4_4.6.1	stringi_1.8.9	stringr_1.6.0
survival_3.8-12	sys_3.4.3	systemfonts_1.3.2
tensorA_0.36.2.1	textshaping_1.0.5	TH.data_1.1-5
threejs_0.3.4	tibble_3.3.1	tidyr_1.3.2
tidyselect_1.2.1	tidyverse_2.0.0	timechange_0.4.0
timeDate_4052.112	tinytex_0.61	tools_4.6.1
tzdb_0.5.0	ucminf_1.2.3	UpSetR_1.4.1
urca_1.3.4	utf8_1.2.6	utils_4.6.1
uuid_1.2.2	V8_8.2.0	vctrs_0.7.3
viridis_0.6.5	viridisLite_0.4.3	visdat_0.6.1
vroom_1.7.1	withr_3.0.3	xfun_0.61
xml2_1.6.0	xtable_1.8-8	xts_0.14.3
yaml_2.3.12	zoo_1.9-1	